

Problem Set 1

Industrial Organization

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I. Conduct analysis in the product space: a monopsonist firm?

Part 1: theory and DGP

i. Solving for the equilibria

In scenario 1, the firm solves

$$\max_{L_t} p_t e^{\nu_t} L_t^\beta - w_t L_t,$$

with FOC

$$w_t = \beta p_t e^{\nu_t} L_t^{\beta-1},$$

which is the labor demand equation in this case. In scenario 2, where the firm knows it will affect wages, we also have to incorporate the $\partial w_t / \partial L_t$ term from the labor supply. The FOC in scenario 2 is

$$\beta p_t e^{\nu_t} L_t^{\beta-1} = w_t + L_t \frac{\partial w_t}{\partial L_t}.$$

We can already see that wages will be lower in this case. Because labor supply is upward-sloping in wages, the equation above will yield the wages of scenario 1 minus a positive term. Taking logs in the labor supply equation, we get

$$\log(L_t) = \alpha + \gamma \log(w_t) + \mu \log(u_t) + \rho \log(u_t) \log(w_t) + \eta_t.$$

This makes it easy to calculate the labor supply elasticity

$$\frac{\partial \log(L_t)}{\partial \log(w_t)} = \frac{w_t}{L_t} \frac{\partial L_t}{\partial w_t} = \gamma + \rho \log(u_t).$$

By the inverse function theorem,

$$\frac{\partial \log(w_t)}{\partial \log(L_t)} = \frac{L_t}{w_t} \frac{\partial w_t}{\partial L_t} = \frac{1}{\gamma + \rho \log(u_t)}.$$

Dividing both sides of the scenario 2 labor demand equation by w_t ,

$$\frac{\beta p_t e^{\nu_t} L_t^{\beta-1}}{w_t} = 1 + \frac{L_t}{w_t} \frac{\partial w_t}{\partial L_t} = 1 + \frac{1}{\gamma + \rho \log(u_t)}.$$

Solving for w_t ,

$$w_t = \beta p_t e^{\nu_t} L_t^{\beta-1} \quad (\text{Scenario 1 Labor Demand})$$

$$w_t = \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1} \beta p_t e^{\nu_t} L_t^{\beta-1}. \quad (\text{Scenario 2 Labor Demand})$$

We can combine both by defining the markdown

$$M = \begin{cases} 1, & \text{in Scenario 1} \\ \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1}, & \text{in Scenario 2} \end{cases}.$$

Then, in both cases, labor market equilibrium (L_t, w_t) is determined by both labor supply and demand, so by the system

$$\begin{cases} w_t = \beta M p_t e^{\nu_t} L_t^{\beta-1} \\ L_t = \exp\left(\alpha + \gamma \log(w_t) + \mu \log(u_t) + \rho \log(u_t) \log(w_t) + \eta_t\right) \end{cases}.$$

If we knew all the parameters $(\beta, \alpha, \gamma, \mu, \rho)$, and we observed $(p_t, u_t, \nu_t, \eta_t)$, we would also know M . Hence, the only unknowns in the two equations would be (L_t, w_t) . We could solve for them by hand by substituting w_t from the labor demand into the labor supply equation, from which we could find a root L_t^* . Then, we just plug it into either equation to compute w_t^* .

1. Wages in Scenario 1 vs. in Scenario 2

Because the markdown term M lies in $(0, 1)$, the labor demand curve is shifted down in Scenario 2. Therefore, equilibrium wages will be lower in Scenario 2.

4. Markdown

By the markdown formula

$$M = \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1},$$

M is increasing in γ , in ρ , and does not depend on μ . It is also increasing in t . Because M is a multiplier, M becoming larger means that wages are being marked down less, hence higher. Here, we rely on $u_t > 1$ to avoid the logs being negative, so unemployment u_t would be defined in absolute terms and not as a fraction.